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Short-term forecasting of electricity demand for the residential sector using weather and social variables

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ABSTRACT

The aim of this study is to provide a precise model for the one-month-ahead forecast of electricity demand in the residential sector. In this study, a total of 20 influential variables are considered including monthly electricity consumption, 14 weather variables, and five social variables. Based on support vector regression and fuzzy-rough feature selection with particle swarm optimization algorithms, the proposed method established a model with variables that relate to the forecast by ignoring variables that inevitably lead to forecasting errors. The proposed forecasting model was validated using historical data from South Korea between January 1991 and December 2012. The first 240 months were used for training and the remaining 24 for testing. The performance was evaluated using MAPE, MAE, RMSE, MBE, and UPA values. Furthermore, it was compared with that obtained from the artificial neural network, autoregressive integrated moving average, multiple linear regression models, and the methods proposed in the previous studies. It was found to be superior for every performance measure considered in this study. The proposed method has an advantage over the previous methods because it automatically determines appropriate and necessary variables for a reliable forecast. It is expected that it can contribute to more accurate forecasting of short-term electricity demand in the residential sector. The ability to accurately forecast short-term electricity demand can assist power system operators and market participants in ensuring sustainable electricity planning decisions and securing electricity supply for the consumers.
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1. Introduction

Short-term electricity demand forecasting plays a significant role in power system planning, including economic scheduling of generating capacity, scheduling of fuel purchases, and power system management (Thatcher, 2007; Shrivastava and Misra, 2008; Kavaklioglu et al., 2009; Mao et al., 2009; Wang et al., 2009; Azadeh et al., 2011; Apadula et al., 2012; Wang, 2012). It is especially obvious that accurate electricity forecasting has great importance to the residential sector, a major contributor to the peak loads in most electricity systems. Overestimating electricity demands misleads planners and wastes resources with expensive expansion plans. Such overestimation also increases operating costs since electricity, unlike other energy sources, cannot be stored on a large scale (Keyno et al., 2009; Qiu, 2013). But underestimation of electricity demands will result in failures and shortages (Kavaklioglu et al., 2009).

Short-term electricity demand forecasting in the residential sector is a complex problem because its rise and fluctuation are caused by the differences in demand from month to month. In addition, consumption is influenced by many nonlinear variables, such as weather conditions, economics, and demographics (Raman and Sunilkumar, 1995; Valor et al., 2001; Altinay and Karagol, 2005; Kavaklioglu et al., 2009; Swan and Ugursal, 2009; Aroonruengsawat and Auffhammer, 2011; Apadula et al., 2012).

Several research studies have been conducted over the last several decades to explore the complex problem of monthly electricity demand forecasting by means of multivariate time series analysis (Quayle and Diaz, 1980; Tserkezos, 1992; Islam et al., 1995; Al-Alawi and Islam, 1996; Al-Zayer and Al-Ibrahim, 1996; Abdel-Aal et al., 1997; Islam and Al-Alawi, 1997; Sailor and Munoz, 1997; Lam, 1998; Yan, 1998; Sailor, 2001; Amato et al., 2005; Hor et al., 2005; Mirasgedis et al., 2006; Pao, 2006; Ruth and Lin, 2006; Lam et al., 2008; Wangpattarapong et al., 2008; Abosedra et al., 2009; Bunnoon et al., 2010; Lam et al., 2010; Pilli-Sihvola et al., 2010; Adam et al., 2011; Chang et al., 2011; Apadula et al., 2012; Bunnoon, 2013; Bunnoon et al., 2013; Chen et al., 2013). Most of the previous studies (Al-Zayer and Al-Ibrahim, 1996; Sailor and Munoz, 1997; Lam, 1998; Hor et al., 2005; Mirasgedis et al., 2006; Lam et al., 2008;

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Wangpattarapong et al., 2008; Apadula et al., 2012) have assumed that the input and output series are stationary and applied statistical models. However, real monthly electricity demand series, as well as variables that may influence the electricity demand series, have been found to be nonstationary (Tserkezos, 1992).

When one or more of the series are nonstationary, it is necessary to consider sophisticated models that are capable of describing the nonlinear input and output series. In order to solve nonlinear time series problems, several research studies applied auto-regressive integrated moving average model (Tserkezos, 1992; Islam and Al-Alawi, 1997) and neural network model-based approaches (Islam et al., 1995; Al-Alawi and Islam, 1996; Pao, 2006; Chang et al., 2011; Bunnoon, 2013; Bunnoon et al., 2013). The accuracy of the resulting models has ranged from 1.42% (Bunnoon et al., 2013) to 10.98% (Islam and Al-Alawi, 1997) in terms of mean absolute percentage error. However, it is difficult to say whether these models have been sufficiently validated, because the evaluation periods of most research studies were within about 10 years (Islam et al., 1995; Abdel-Aal et al., 1997; Islam and Al-Alawi, 1997; Adam et al., 2011; Chang et al., 2011; Bunnoon, 2013; Bunnoon et al., 2013; Chen et al., 2013). To our knowledge, there have been no research studies examining the method that can develop a model by determining appropriate and necessary variables as well as handling nonlinear time series problems for a reliable forecast of monthly electricity demand in the residential sector.

The aim of this study is to provide a precise model for one-month-ahead forecast of electricity demand in the residential sector. Based on support vector regression and fuzzy-rough feature selection with particle swarm optimization algorithms, the proposed method automatically develops a forecasting model with variables that relate to the electricity demand series by ignoring variables that inevitably lead to forecasting errors. To evaluate the forecasting performance of the proposed method, we performed a comprehensive comparison of the prediction performance of the proposed method versus that of the artificial neural network, auto-regressive integrated moving average, multiple linear regression models, and the methods proposed in the previous studies. A data set covering the period from January 1991 to December 2012 was collected from South Korea and used for training and testing experiments.

In Section 2, a comprehensive review of the literature is presented. In Section 3, the data set and pre-processing and some materials on the proposed methodology are described. In Section 4, a discussion and analysis of the experimental results are presented. Section 5 contains conclusions and suggestions for future research.

2. Related works

According to the Annual Energy Outlook 2014 by the U.S. Energy Information Administration, although the growth in electricity use has slowed down, it is still forecasted to increase by 29% from 2012 to 2040, with the residential sector accounting for the largest portion of energy consumption (US Energy Information Administration, 2014). Depending on the country, the residential sector accounted for anywhere between 16% and 50% of energy consumption, averaging approximately 30% worldwide (Swan and Ugursal, 2009). This considerable amount of the energy consumption in the residential sector shows that a thorough understanding is required to prepare for and help guide consumption in an increasingly energy-conscious world, particularly with regard to supply, efficient use, and effects of consumption (Swan and Ugursal, 2009).

As part of this point of view, several research studies have explored this complex problem of monthly electricity demand forecasting by means of multivariate time series analysis (Quayle and Diaz, 1980; Tserkezos, 1992; Islam et al., 1995; Al-Alawi and Islam, 1996; Al-Zayer and Al-Ibrahim, 1996; Abdel-Aal et al., 1997; Islam and Al-Alawi, 1997; Sailor and Munoz, 1997; Lam, 1998; Yan, 1998; Sailor, 2001; Amato et al., 2005; Hor et al., 2005; Mirasgedis et al., 2006; Pao, 2006; Ruth and Lin, 2006; Lam et al., 2008; Wangpattarapong et al., 2008; Abosedra et al., 2009; Bunnoon et al., 2010; Lam et al., 2010; Pilli-Sihvola et al., 2010; Adam et al., 2011; Chang et al., 2011; Apadula et al., 2012; Bunnoon, 2013; Bunnoon et al., 2013; Chen et al., 2013). This chapter comprehensively reviews the variables which appear to have been considered in the literature. To date, many nonlinear variables, such as weather conditions, economics, and demographics, were found to have influence on electricity demand series (Raman and Sunilkumar, 1995; Valor et al., 2001; Altinay and Karagol, 2005; Kavaklioglu et al., 2009; Swan and Ugursal, 2009; Aroonruengsawat and Auffhammer, 2011; Apadula et al., 2012). Table 1 summarizes some of the many representative variables related to the weather conditions, economics, and demographics. The comprehensive list in Table 1 was developed from 28 studies.

The variables that were considered no more than twice in the literature were not included in Table 1, so research studies by Mirasgedis et al. (2006) and Apadula et al. (2012) were not included the list. For weather variables, the variables excluded from Table 1 were *absolute maximum temperature* and *average maximum relative humidity* (Islam et al., 1995; Islam and Al-Alawi, 1997), *maximum relative humidity* (Al-Alawi and Islam, 1996), *comfort index* (Al-Alawi and Islam, 1996; Islam and Al-Alawi, 1997), *temperature at peak load* and *relative humidity at peak load* (Islam and Al-Alawi, 1997), *cooling radiation-days* (Lam, 1998), *clo* (Yan, 1998), *cloud cover* (Yan, 1998; Apadula et al., 2012), *annual trend heating degree days* and *annual trend cooling degree days* (Amato et al., 2005), *total number of monthly heating degree days* and *total number of monthly cooling degree days* (Mirasgedis et al., 2006), *dry-bulb temperature* and *wet-bulb temperature* (Lam et al., 2008; Bunnoon et al., 2010), *clearness index* (Lam et al., 2008, 2010), *rainy time* and *average air pressure* (Chang et al., 2011), and *heating degree months* and *cooling degree months* (Apadula et al., 2012).

For social variables, the variables excluded from Table 1 were *number of consumers connected* (Al-Alawi and Islam, 1996; Islam and Al-Alawi, 1997), *household size* (Lam, 1998), *household income* (Lam, 1998; Wangpattarapong et al., 2008), *number of non-working days during the month* (Mirasgedis et al., 2006), national income (Pao, 2006), *number of houses* and *monthly number of air conditioners sold* (Wangpattarapong et al., 2008), *real import* (Abosedra et al., 2009), *weighted calendar variable* (Apadula et al., 2012), and *gross value of export-import*, *industry increasing value*, and *social retail sales of consumer goods* (Chen et al., 2013). Although some of the selected variables may inevitably lead to forecasting errors, most of the previous studies have selected such variables based on human intuition and expertise in this subject area or based on validation results of one-to-one correlations between the variables and electricity demand series to develop forecasting models (Islam et al., 1995; Al-Zayer and Al-Ibrahim, 1996; Islam and Al-Alawi, 1997; Sailor and Munoz, 1997; Lam, 1998; Hor et al., 2005; Lam et al., 2008; Wangpattarapong et al., 2008; Apadula et al., 2012).

Most of the previous studies have assumed that certain inputs among various weather variables and social variables have an impact on the electricity demand series, and therefore they fed these inputs to develop their models. In some cases, although some selected variables may inevitably lead to forecasting errors, the variables were selected based on human intuition and expertise in this subject area or based on validation results of one-to-one correlations between the variables and electricity demand series (Islam et al., 1995; Al-Zayer and Al-Ibrahim, 1996; Islam and Al-Alawi, 1997; Sailor and Munoz, 1997; Lam, 1998; Hor et al., 2005; Lam et al., 2008; Wangpattarapong et al., 2008; Apadula et al., 2012). None have examined the method that can develop a model

Table 1
Weather and social variables considered in the literature.

Variable name	Quayle and Diaz (1980)	Tserkezos (1992)	Islam et al. (1995)	Al-Alawi and Islam (1996)	Al-Zayer and Al-Ibrahim (1996)	Abdel-Aal et al. (1997)	Islam and Al-Alawi (1997)	Sailor and Munoz (1997)	Lam (1998)	Yan (1998)	Sailor (2001)	Amato et al. (2005)	Hor et al. (2005)
Weather variables													
Mean temperature				○		○		○		○			
Maximum temperature		○	○				○						
Minimum temperature													
Relative humidity			○	○		○	○	○					○
Wind speed			○	○			○	○			○		○
Rain fall			○				○						○
Daylight time			○	○			○					○	○
Global solar radiation			○	○		○	○						
Cooling degree days	○							○			○	○	○
Heating degree days								○	○		○	○	○
Degree days				○	○		○						
Vapor pressure			○				○			○			
Enthalpy latent days								○	○		○		○
Social variables													
Real gross domestic product						○							○
Industrial production index													
Population						○							
Consumer price index													
Real electricity price		○							○			○	

Variable name	Pao (2006)	Ruth and Lin (2006)	Lam et al. (2008)	Wangpattarapong et al. (2008)	Abosedra et al. (2009)	Bunnoon et al. (2010)	Lam et al. (2010)	Pilli-Sihvola et al. (2010)	Adam et al. (2011)	Chang et al. (2011)	Bunnoon (2013)	Bunnoon et al. (2013)	Chen et al., 2013
Weather variables													
Mean temperature					○	○			○	○	○	○	○
Maximum temperature						○					○	○	
Minimum temperature						○					○	○	
Relative humidity				○	○	○			○	○	○		
Wind speed			○	○		○	○			○	○		
Rain fall				○		○				○	○		
Daylight time		○							○	○			
Global solar radiation			○				○						
Cooling degree days		○						○					
Heating degree days		○		○				○					
Degree days													
Vapor pressure													
Enthalpy latent days													
Social variables													
Real gross domestic product	○								○				
Industrial production index						○					○	○	
Population	○			○									
Consumer price index	○					○					○	○	○
Real electricity price			○										

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Table 2
Summary statistics for the residential electricity consumption and 19 variables.

No.	Variable name (Unit)	Mean	Std. deviation	Minimum	Maximum
1	Electricity consumption (MWh)	3,478,541.87	1,167,190.45	1,492,690.00	6,256,403.00
2[a]	Mean temperature (°C)	13.40	9.11	−4.95	28.64
3[a]	Maximum temperature (°C)	18.20	8.80	0.46	33.31
4[a]	Minimum temperature (°C)	7.83	9.44	−9.84	23.77
5[a]	Relative humidity (%)	63.47	8.50	45.04	82.66
6[a]	Wind speed (m/s)	2.44	0.31	1.67	3.26
7[a]	Rainfall (mm)	120.79	140.89	3.22	804.15
8[a]	Daylight time (h)	175.71	36.97	55.88	256.73
9[a]	Global solar radiation (0.01 MJ/m^2)	38,151.02	11,000.99	16,889.65	63,613.40
10[a]	Cooling degree days (degree-days)	10.47	24.46	0.00	147.09
11[a]	Heating degree days (degree-days)	201.07	211.24	0.00	714.58
12[a]	Total number of cooling degree days (degree-days)	10.29	24.49	0.00	147.09
13[a]	Total number of heating degree days (degree-days)	200.88	211.42	0.00	714.58
14[a]	Vapor pressure (0.1 hPa)	119.91	75.94	22.04	291.87
15[a]	Air pressure (0.1 hPa)	10,161.31	64.81	10,047.08	10,271.27
16	Real gross domestic product (billions of won)	191,076.33	44,693.43	103,141.20	273,719.10
17	Industrial production index	238.98	105.79	85.32	446.24
18	Population (millions)	47,766.42	2042.62	43,296.00	50,948.27
19	Consumer price index	165.60	36.28	100.00	228.97
20	Real electricity price (won/kWh)	47.32	5.88	38.10	69.53

[a] Indicates that it was calculated by population-weighted average of the nine regions they represent. Monthly data were interpolated assuming smooth changes from the quarterly data (real gross domestic product and population).

by determining appropriate and necessary variables as well as handling nonlinear time series problems for a reliable forecast of monthly electricity demand in the residential sector.

3. Methodology

3.1. Data set and pre-processing

There are 20 related variables (as shown in Table 2) including monthly electricity consumption, 14 weather variables, and five social variables (economic and demographic factors). The time period of data collection is January 1991 to December 2012; in other words, a total 264 monthly data are present. In this study, the monthly residential electricity consumption series and variables that may influence the electricity demand series are collected from statistics published by the Korea Meteorological Administration, Korean Statistical Information System, and Korea Energy Statistics Information System. Although many more than those variables in Table 2 were reviewed in the literature, the variables were screened and excluded first based on whether they were collectible or not during the entire period. Then, the variables were selected from each category based on their representativeness and frequency of occurrence in the literature.

In South Korea, monthly weather data were obtained from nine regional representative monitoring stations. Each monitoring station was associated with a climate region, and these climate regions represent areas of similar climates and are therefore somewhat arbitrary. The data for each monitoring station include averages of daily temperature, maximum and minimum temperatures, relative humidity, wind speed, rainfall, daylight time, global solar radiation, cooling and heating degree days, total numbers of cooling and heating degree days, vapor pressure, and air pressure. Then, these 14 variables were calculated by the population-weighted average of the nine regions they represent. The reason why population has been selected as a weighting factor is that climate influences electric consumption through people's response to weather. That is, depending on how hot or cold the weather is, people will increase or decrease their use of electric heating appliances or air conditioners. Thus, the higher the population, the higher the influence of weather conditions on electricity demand (Valor et al., 2001). In other words, regional electricity demand is highly correlated with population (Sailor and Munoz, 1997). Table 2 presents the summary statistics for residential electricity consumption and 19 variables in South Korea from January 1991 to December 2012.

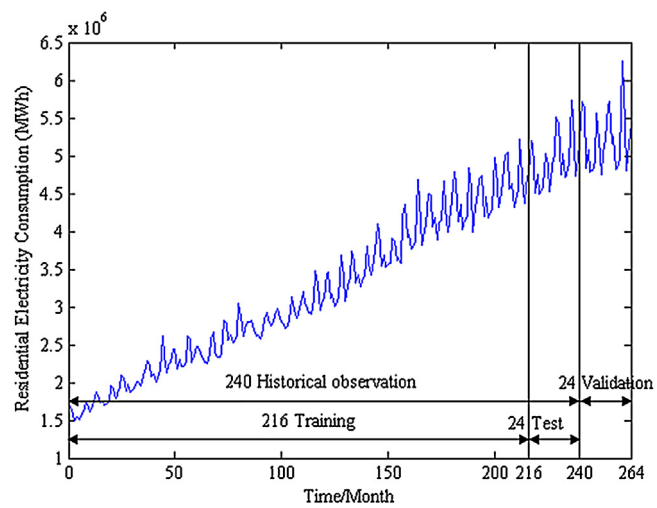

Fig. 1. Monthly residential electricity consumption from January 1991 to December 2012.

The time series of residential electricity consumption is plotted in Fig. 1 from January 1991 to December 2012, a total of 264 months of data. The series presents complex monthly and seasonal trends and strong fluctuations. These data exhibit a significant and persistent upward trend until the end of 2012, mainly associated with the country's economic and demographic growth. In addition, a cyclic annual pattern, which is correlated with global climate changes throughout the year, is superimposed over these monthly and seasonal trends. The amplitude of the collected values also reflects the different number of days, working days, and holidays included in each month.

3.2. Analysis steps

This study was conducted according to the procedure outlined in Fig. 2. In Step 1, support vector regression (SVR) and fuzzy-rough feature selection with particle swarm optimization (PSO)

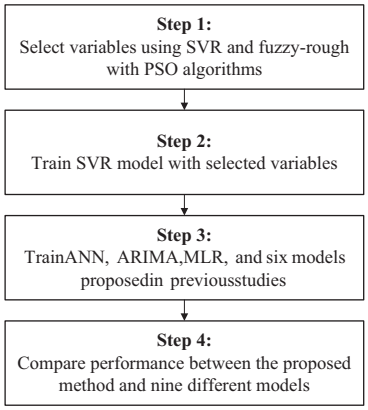

Fig. 2. Outline of procedure of the proposed method.

algorithms were applied to select the most relevant variables from the 19 input variables. As a result, the number of multi-dimensional input variables was reduced. In Step 2, a forecasting model was developed by training the support vector regression model with the variables selected in the first step. In Step 3, the artificial neural network (ANN), auto-regressive integrated moving average (ARIMA), multiple linear regression (MLR) models, and the methods proposed in the previous studies were trained. In Step 4, the forecasting performance of each of the 10 models was compared.

3.2.1. *Fuzzy-rough feature selection with particle swarm optimization*

In this study, the fuzzy-rough feature selection method, proposed by Moaref and Naeini (2013), is modified. This method is based on the PSO algorithm that utilizes fuzzy dependency degree as the evaluation measure. The algorithm initiates by generating N numbers of particles randomly, equaling half the variables. Each particle indicates a subset of input variables. The best particles are obtained in all iterations and are called global bests. Then, personal best of each particle and the best immediate neighbors for each particle are obtained. The velocity of the particle is updated and the position update rule can be formulated. The fitness of the particle is calculated and the termination condition is checked. Details of the fuzzy-rough algorithm can be found in Cornelis and Jensen (2008), Cornelis et al. (2008), and Jensen and Shen (2009). In addition, details of the PSO algorithm can be found in Wang et al. (2006, 2007).

3.2.2. *Support vector regression (SVR)*

As indicated in previous studies, monthly electricity demand has a cyclic trend caused by the difference in demand from month to month and season to season. However, the application of the SVR model, which is capable of dealing with cyclic trend time series, has yet to be explored (Hong, 2011). In real applications in time series prediction, the SVR algorithm has received extensive attention as a learning algorithm offering many theoretical advantages (Müller et al., 1997; Crone et al., 2006). The variables obtained in the feature selection step were passed to the SVR forecasting model for validation. A proper parameter setting can improve the forecasting performance of an SVR. With an RBF kernel, there are two parameters to be determined in the SVR model: the regularization parameter, C, and the kernel parameter, gamma. Proper selection of C and gamma values could improve the forecasting performance of the SVR. The "grid-search" approach proposed by Hsu et al. (2003) and Wang et al. (2005) is an alternative to finding the best C and gamma when using an RBF kernel function. The grid search was performed as follows (Hsu et al., 2003): First, we selected a grid space with C (10^{-2}, 10^{-1}, ..., 10^{4}, 10^{5}) and gamma (10^{-5}, 10^{-4}, ..., 10^{0}, 10^{1}). Then, for each pair of C and gamma in this space, the RMSE was calculated for the test period. Finally, the pair C and gamma that yielded the smallest value of the RMSE was chosen, and that pair was used to train the entire training set and generate the final forecasting model.

4. Results and discussion

The performance of the proposed method is evaluated using the collected data set. The results obtained by the proposed method were compared with the results obtained by the artificial neural network, auto-regressive integrated moving average, multiple linear regression models, and the methods proposed in the previous studies. The experimental results are summarized in Tables 3 and 4 and illustrated in Figs. 3 and 4. The first step of the procedure consisted of selecting the most relevant subset of variables from the set of 19 candidate variables. Based on support vector regression and fuzzy-rough feature selection with particle swarm optimization algorithms, 11 variables including monthly electricity consumption, 8 weather variables, and 2 social variables were utilized for modeling the monthly electricity consumption (see Table 3).

The performance was evaluated using mean absolute percentage error (MAPE), mean absolute error (MAE), root mean squared error (RMSE), mean bias error (MBE), and unpaired peak accuracy (UPA) values. MAE and RMSE values allow a term-by-term comparison of the actual deviation between the forecasted value and actual value. Since the MAE and RMSE are based on absolute errors, there is no absolute criterion for a "good" value for either of them. All that can be inferred from them is that the smaller the value of MAE or RMSE, the closer the predicted values to the actual values. The MAPE, however, is scale independent, since it is based on relative errors; hence, it is more meaningful (Makridakis, 1993). Positive and negative values of MBE represent overestimated and underestimated forecasted values, with a small value of MBE being desirable. The UPA value gives a measure of the model's ability to capture peak demand, but does not pair the model forecasts with actual value in time or space. The performance statistics of the proposed model and nine different models are given in Table 4. The proposed method achieves the highest performance among the 10 models in terms of MAPE, MAE, RMSE, and MBE values. At the peak point, the proposed method was ranked in the second position, yielding −5.89 in terms of UPA value. In other words, it means that the residential electricity demand at the peak point was underestimated by the proposed method with 5.89% error.

Graphical comparison of the actual values (blue line) to the forecasted values (red line) by the proposed method from January 2011 to December 2012 is presented in Fig. 3. The time series of residential electricity consumption is also plotted in Fig. 3. The series presents a complex monthly and seasonal trends and strong fluctuations. These data exhibit a significant and persistent upward trend until the end of 2012, mainly associated with the economic and demographic growth in the country. In addition, a cyclic annual pattern, which is correlated with the global climate changes throughout the year, is superimposed over these monthly and seasonal trends. The amplitude of the collected values also reflects the different number of days, working days, and holidays included in each month. The forecasted values (red line) by the proposed model are closer to the actual values. From the results, one could conclude that the forecasted values show relatively good agreement with the actual values and that the proposed method is feasible and reliable.

Fig. 4 shows graphical comparisons of the actual values to the values forecasted by the proposed method and the other five models, which yield less than 5% in terms of MAPE from January 2011 to December 2012. From our analysis of the empirical results, it appears that the proposed model performed better than any of the

Table 3
Selected variables using the proposed method for modeling the monthly electricity demand.

No.	Variable name (Unit)	No.	Variable name (Unit)
1[a]	Mean temperature (°C)	6[a]	Cooling degree days (degree-days)
2[a]	Maximum temperature (°C)	7[a]	Total number of cooling degree days (degree-days)
3[a]	Wind speed (m/s)	8[a]	Vapor pressure (0.1 hPa)
4[a]	Daylight time (h)	9	Consumer price index
5[a]	Global solar radiation (0.01 MJ/m^2)	10	Real electricity price (won/kWh)

[a] Indicates that it was calculated by population-weighted average of the nine regions they represent.

Table 4
Comparison of 10 different models.

Method	MAPE	MAE	RMSE	MBE	UPA
Proposed method	**2.13**	**113,870.67**	**149,287.54**	**45,695.11**	−5.89
Artificial neural network (ANN)	9.92	522,503.25	633,073.65	500,754.67	14.03
Auto-regressive integrated moving average (ARIMA)	4.71	242,005.30	261,671.62	224,571.82	**−2.57**
Multiple linear regression (MLR)	7.15	361,170.52	426,779.00	300,159.62	−7.14
Proposed by Islam and Al-Alawi (1997)	3.96	203,109.63	241,167.57	160,884.52	−6.88
Proposed by Hor et al. (2005) – Model 1	2.78	152,485.53	220,244.03	−95,728.85	−12.05
Proposed by Hor et al. (2005) – Model 2	3.30	175,170.65	231,332.24	−131.75	−9.51
Proposed by Hor et al. (2005) – Model 3	4.53	244,468.27	301,318.18	−229,422.22	−14.06
Proposed by Pao et al. (2006)	6.09	331,167.88	417,682.95	−331,167.88	−19.61
Proposed by Bunnoon et al. (2010)	7.70	411,642.53	520,294.26	−352,916.51	−14.83

Note. The values were rounded to four decimals places.
The lowest MAPE, MAE, RMSE, MBE, and UPA values are in bold.

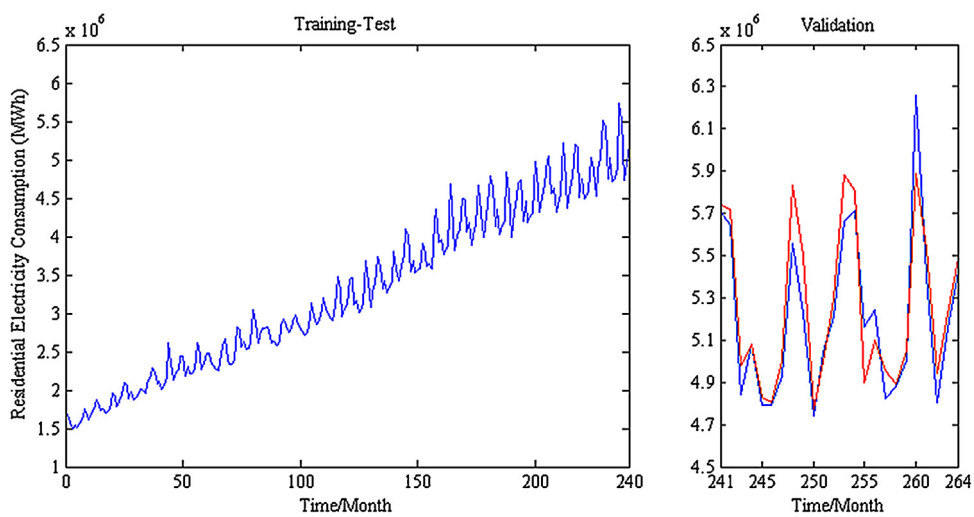

Fig. 3. Forecasting results using the proposed method. (For interpretation of the references to color in the text, the reader is referred to the web version of this article.)

other models, in that it yielded the smallest MAPE, MAE, RMSE, and MBE values. At this point, we are interested in doing a more in-depth examination of whether the proposed method is superior to the other models with regard to forecasting of electricity demand in the residential sector. We are also interested in determining whether the rankings of the other models with respect to the four aforementioned measures (MAPE, MAE, RMSE, and MBE) have any particular significance in this regard.

In order to test whether the proposed method is superior to the other models, we applied the two-tailed Wilcoxon matched-pairs signed-ranks test with respect to these four measures. This test was chosen because it is a non-parametric method and it imposes no restriction on the underlying distributions in the data. In addition, it is not performed on the magnitudes of the values of the variables but on their signs and ranks; thus, it is not influenced by outlier data points. It is one of the most widely adopted tests used in evaluating the predictive capabilities of different models to see whether there is a statistically significant difference between them (Pollock et al., 2005; Lu et al., 2009). Details of the Wilcoxon signed-ranks test can be found in Diebold and Mariano (1995) and Pollock et al. (2005).

The null hypothesis of the two-tailed Wilcoxon signed-ranks test is that the difference between the values of each of the MAPE, MAE, RMSE, and MBE values for the two models being compared is zero. If the performance of one model considerably surpasses the other, those differences will be significantly different from zero. The null hypothesis is rejected (meaning that the values of the four measures for the two models are significantly different) when the p-value is below a pre-established level of significance. In this study, the minimum significance level for rejecting a null hypothesis was set at 5%. As a result, the values of the four measures for the proposed method are significantly different from those for the other models. Since the proposed method yielded the smallest MAPE, MAE, RMSE, and MBE values in this study, as well as the best scores on the two-tailed Wilcoxon signed-ranks test for those two measures, we can conclude that the prediction performance of the proposed model is significantly better than that of the other models

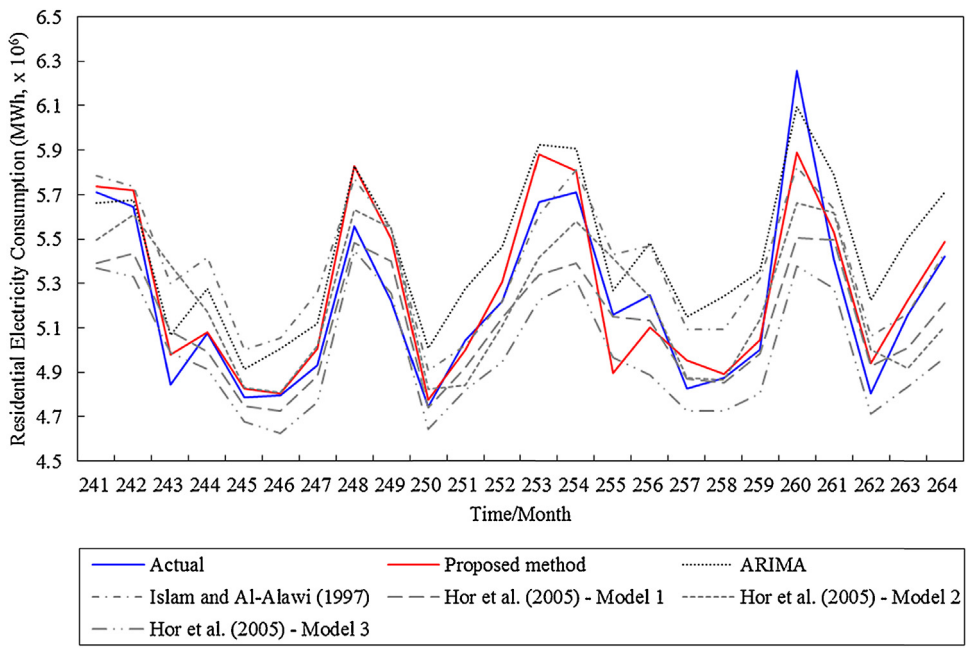

Fig. 4. Actual and forecasted values from six different models.

in regard to forecasting electricity consumption in the residential sector.

5. Conclusion

This study proposed a method to provide a precise model for the one-month-ahead forecast of electricity demand in the residential sector. The support vector regression and fuzzy-rough feature selection with particle swarm optimization algorithms identified 10 variables from a set of 19 candidate variables that may influence the electricity demand series. Then, the support vector regression algorithm was used to construct a forecasting model of electricity demand based on the electricity consumption series and 10 selected variables. The use of the proposed method allowed us to reduce the dimensionality of the input variables fed to the support vector regression model by eliminating variables that may lead to forecasting errors. With fewer—and more relevant—input variables, the support vector regression model was better able to describe the rise and fluctuation of electricity demand series in the residential sector. Consequently, the proposed method was able to more accurately forecast the electricity demand in the residential sector.

In summary, several interesting findings were made in this study. First, prior to this study there was relatively little understanding of a variable selection method that significantly contributes the forecasting of electricity demand in the residential sector. This study proposed the method that can identify relevant variables for forecasting the electricity demand in the residential sector with comprehensive consideration of variables that may influence the electricity demand series. The results imply that the electricity demand in the residential sector can be forecasted with reliable accuracy one month ahead based upon the values of the variables selected by the proposed method. Second, the forecasting performance obtained using the proposed method was compared with that obtained using the artificial neural network, auto-regressive integrated moving average, multiple linear regression models, and the methods proposed in the previous studies. The results of the comparison confirmed the proposed method is the best predictor of the monthly electricity demand in the residential sector, as it yields smaller errors than the other models. The proposed method has an advantage over the previous methods because it automatically determines appropriate and necessary variables for a reliable forecast.

The proposed method has an advantage over the previous methods because it automatically determines appropriate and necessary variables for a reliable forecast. It is expected that the proposed model can contribute to more accurate forecasting of short-term electricity demand in the residential sector. The ability to accurately forecast short-term electricity demand can assist power system operators and market participants in ensuring sustainable electricity planning decisions and secure electricity supply to the consumers. This highly accurate forecasting method also has great potential for solving other forecasting problems in the construction industry.

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